

## “Phase I” method for initial feasible point (0.25 pt)

We consider the original constraints:

$$\begin{aligned}(\text{C1}) \quad & x - 2y + 2 \geq 0, \\(\text{C2}) \quad & -x - 2y + 6 \geq 0, \\(\text{C3}) \quad & -x + 2y + 2 \geq 0, \\(\text{C4}) \quad & x \geq 0, \\(\text{C5}) \quad & y \geq 0.\end{aligned}$$

Given the starting point

$$x_0 = \begin{pmatrix} 1 \\ 2.5 \end{pmatrix},$$

we first check feasibility:

$$\begin{aligned}\text{C1: } & 1 - 2 \cdot 2.5 + 2 = -2 \quad (\text{violated by } 2), \\ \text{C2: } & -1 - 2 \cdot 2.5 + 6 = 0 \quad (\text{active}), \\ \text{C3: } & -1 + 2 \cdot 2.5 + 2 = 6 \quad (> 0), \\ \text{C4: } & 1 \geq 0, \quad \text{C5: } 2.5 \geq 0.\end{aligned}$$

Only Constraint (C1) is violated.

### Phase I reformulation

Introduce the slack variable  $s \geq 0$  to account for infeasibility and modify the constraints to follow the convention used in the lecture:

$$\begin{aligned}\min_{x,y,s} \quad & s \\ \text{s.t.} \quad & -x + 2y - 2 \leq s, \\ & x + 2y - 6 \leq 0, \\ & x - 2y - 2 \leq 0, \\ & -x \leq 0, \quad -y \leq 0, \quad s \geq 0.\end{aligned}$$

Choose  $s_0 = 2 = \max(0, g_1(x_0))$  so that Constraint (C1) becomes satisfied at the starting point:

$$-1 + 2 \cdot 2.5 - 2 = 2.$$

Thus  $(x_0, y_0, s_0) = (1, 2.5, 2)$  is feasible for the Phase I problem.

### Solving Phase I

Because the original problem is feasible, the minimum of Phase I is  $s^* = 0$ , attained at any point satisfying (C1)–(C5).

A suitable feasible point is found by intersecting (C1) and (C2):

$$\begin{cases} -x + 2y - 2 = 0, \\ x + 2y - 6 = 0. \end{cases}$$

Solve the system:

$$\begin{aligned}x &= 2y - 2, \\ (2y - 2) + 2y - 6 &= 0 \Rightarrow 4y - 8 = 0 \Rightarrow y = 2,\end{aligned}$$

$$x = 2 \cdot 2 - 2 = 2.$$

Thus the candidate point is:

$$(x, y, s) = (2, 2, 0).$$

Check feasibility:

$$\text{C1: } 2 - 2 \cdot 2 + 2 = 0 \geq 0,$$

$$\text{C2: } -2 - 2 \cdot 2 + 6 = 0 \geq 0,$$

$$\text{C3: } -2 + 2 \cdot 2 + 2 = 4 \geq 0,$$

$$\text{C4: } 2 \geq 0, \quad \text{C5: } 2 \geq 0,$$

$$s = 0.$$

Thus, a valid feasible point for the original problem, produced by the Phase I method, is

$$x_{\text{feasible}} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

which satisfies all original constraints and can be used to start the active-set method.

## ”Big M” method for initial feasible point (Bonus 0.5 pt)

We introduce the variable

$$z := \begin{pmatrix} x \\ y \\ s \end{pmatrix} \in \mathbb{R}^3$$

and rewrite the Big M formulation as a quadratic program of the form

$$\min_z \frac{1}{2} z^\top Q z + q^\top z + r \quad \text{subject to} \quad Bz \leq b.$$

The Big M objective is

$$F_M(x, y, s) = (x - 1)^2 + (y - 2.5)^2 + 100s,$$

so

$$Q = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad q = \begin{pmatrix} -2 \\ -5 \\ 100 \end{pmatrix}, \quad r = 7.25.$$

The constraints

$$-x + 2y - 2 \leq s,$$

$$x + 2y - 6 \leq 0,$$

$$x - 2y - 2 \leq 0,$$

$$-x \leq 0, \quad -y \leq 0, \quad s \geq 0$$

are written as

$$Bz \leq b, \quad B \in \mathbb{R}^{6 \times 3}, \quad b \in \mathbb{R}^6,$$

with rows

$$B = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 1 & -2 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

The starting point is

$$z_0 = \begin{pmatrix} 1 \\ 2.5 \\ 2 \end{pmatrix},$$

for which

$$Bz_0 - b = \begin{pmatrix} 0 \\ 0 \\ -6 \\ -1 \\ -2.5 \\ -2 \end{pmatrix} \leq 0,$$

so  $z_0$  is feasible and constraints 1 and 2 are active:

$$\mathcal{A}_0 = \{1, 2\}.$$

### Active-set method for the Big M QP

For a given active set  $\mathcal{A}_k$ , let  $B_k$  and  $b_k$  collect the corresponding rows of  $B$  and entries of  $b$ . The active-set subproblem is

$$\min_z \frac{1}{2} z^\top Q z + q^\top z + r \quad \text{s.t.} \quad B_k z = b_k,$$

whose KKT system is

$$\begin{pmatrix} Q & B_k^\top \\ B_k & 0 \end{pmatrix} \begin{pmatrix} z_{k+1}^{\text{cand}} \\ \lambda_{k+1, \mathcal{A}_k} \end{pmatrix} = \begin{pmatrix} -q \\ b_k \end{pmatrix}.$$

This gives a candidate point  $z_{k+1}^{\text{cand}}$ ; the search direction is  $p_k = z_{k+1}^{\text{cand}} - z_k$ , and we choose a step length  $\alpha_k \in (0, 1]$  so that  $z_k + \alpha_k p_k$  remains feasible. If  $\alpha_k < 1$ , a new constraint becomes active. If  $p_k = 0$ , we inspect the multipliers and possibly drop a constraint with negative multiplier.

**Iteration 0.** Active set:  $\mathcal{A}_0 = \{1, 2\}$ , so

$$B_0 = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \end{pmatrix}, \quad b_0 = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

The KKT system

$$\begin{pmatrix} Q & B_0^\top \\ B_0 & 0 \end{pmatrix} \begin{pmatrix} z_1^{\text{cand}} \\ \lambda_{1, \mathcal{A}_0} \end{pmatrix} = \begin{pmatrix} -q \\ b_0 \end{pmatrix}$$

yields

$$z_1^{\text{cand}} = \begin{pmatrix} 81 \\ -37.5 \\ -158 \end{pmatrix}, \quad \lambda_{1,1} = 100, \quad \lambda_{1,2} = -60.$$

The search direction is

$$p_0 = z_1^{\text{cand}} - z_0 = \begin{pmatrix} 80 \\ -40 \\ -160 \end{pmatrix}.$$

The inactive constraints along  $z(\alpha) = z_0 + \alpha p_0$  satisfy

$$\begin{pmatrix} 160\alpha - 6 \\ -80\alpha - 1 \\ 40\alpha - 2.5 \\ 160\alpha - 2 \end{pmatrix} \leq 0,$$

which gives

$$\alpha_0 \leq \min \left\{ \frac{6}{160}, \frac{2.5}{40}, \frac{2}{160} \right\} = \frac{1}{80}.$$

The blocking constraint is 6, so  $\alpha_0 = 1/80$  and

$$z_1 = z_0 + \alpha_0 p_0 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}.$$

At  $z_1$ ,  $Bz_1 - b = (0, 0, -4, -2, -2, 0)^\top$ , so

$$\mathcal{A}_1 = \{1, 2, 6\}.$$

**Iteration 1.** With  $\mathcal{A}_1 = \{1, 2, 6\}$ ,

$$B_1 = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}.$$

Solving

$$\begin{pmatrix} Q & B_1^\top \\ B_1 & 0 \end{pmatrix} \begin{pmatrix} z_2^{\text{cand}} \\ \lambda_{2,\mathcal{A}_1} \end{pmatrix} = \begin{pmatrix} -q \\ b_1 \end{pmatrix}$$

gives

$$z_2^{\text{cand}} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = z_1, \quad \lambda_{2,1} = \frac{5}{4}, \quad \lambda_{2,2} = -\frac{3}{4}, \quad \lambda_{2,6} = \frac{395}{4}.$$

Hence  $p_1 = 0$ . Since one multiplier is negative ( $\lambda_{2,2} < 0$ ), constraint 2 is removed from the active set:

$$\mathcal{A}_2 = \{1, 6\}, \quad z_2 := z_1.$$

**Iteration 2.** Now

$$B_2 = \begin{pmatrix} -1 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

The KKT system

$$\begin{pmatrix} Q & B_2^\top \\ B_2 & 0 \end{pmatrix} \begin{pmatrix} z_3^{\text{cand}} \\ \lambda_{3,\mathcal{A}_2} \end{pmatrix} = \begin{pmatrix} -q \\ b_2 \end{pmatrix}$$

yields

$$z_3^{\text{cand}} = \begin{pmatrix} \frac{7}{5} \\ \frac{17}{10} \\ 0 \end{pmatrix}, \quad \lambda_{3,1} = \frac{4}{5}, \quad \lambda_{3,6} = \frac{496}{5}.$$

The search direction is

$$p_2 = z_3^{\text{cand}} - z_2 = \begin{pmatrix} -\frac{3}{5} \\ -\frac{3}{10} \\ 0 \end{pmatrix}.$$

Along  $z(\alpha) = z_2 + \alpha p_2$ , the inactive constraints satisfy

$$\begin{pmatrix} -\frac{6}{5}\alpha \\ -4 \\ \frac{3}{5}\alpha - 2 \\ \frac{3}{10}\alpha - 2 \end{pmatrix} \leq 0$$

for all  $\alpha \in [0, 1]$ . Thus no new constraint blocks the step and we take  $\alpha_2 = 1$ :

$$z_3 = z_2 + p_2 = \begin{pmatrix} \frac{7}{5} \\ \frac{17}{10} \\ 0 \end{pmatrix}.$$

At  $z_3$ ,  $Bz_3 - b = (0, -\frac{6}{5}, -4, -\frac{7}{5}, -\frac{17}{10}, 0)^\top$ , so

$$\mathcal{A}_3 = \{1, 6\}.$$

**Iteration 3 (optimality).** Solving the KKT system again with  $\mathcal{A}_3 = \{1, 6\}$  returns

$$z_4^{\text{cand}} = z_3, \quad \lambda_{4,1} = \frac{4}{5}, \quad \lambda_{4,6} = \frac{496}{5},$$

so  $p_3 = 0$  and all multipliers of active constraints are nonnegative. Therefore the KKT conditions are satisfied and the active-set method terminates at

$$(x^*, y^*, s^*) = \left( \frac{7}{5}, \frac{17}{10}, 0 \right) = (1.4, 1.7, 0).$$

This matches the solution we found in the first exercise as expected.